Grammar-Aware Fuzzing of PostgreSQL, and Beyond

CS-5187 Software Analysis, Term Project A3

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November 2026

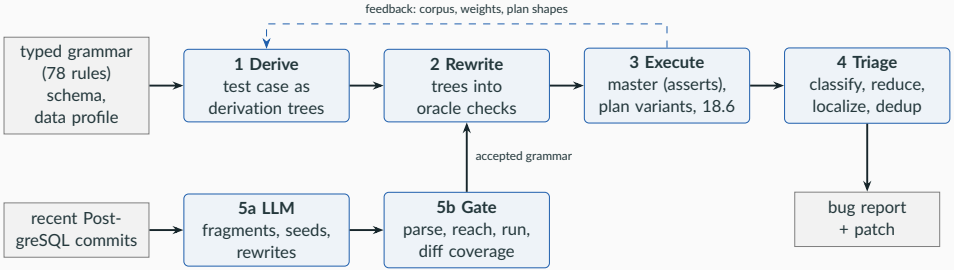
IIT Guwahati

Target and problem

- **Target**: PostgreSQL master built with **assertions**; reference: release 18.6.
- **Bug classes**: crash/assertion, wrong result, internal error, hang.
- A wrong result leaves no error and no log line: the tool needs a test oracle that knows the right answer.

Our idea
Generate SQL as typed derivation trees. Rewrite each tree into equivalent queries, run it under other plans and on the last release. Any disagreement is a bug.

Arbor: architecture and techniques



- **Typed grammar** (own DSL, 78 rules): 99.3% valid statements (random mutation 26%); tree kept for every query.
- **Oracles**: crash/assertion, internal error, hang; 18 **plan variants**; **metamorphic tree rewrites** (TLP, NoREC, $e \to (\mathrm{SELECT}\ e)$); **cross-version**; **CERT** estimates.
- **Triage**: reproduce $\times 3$, reduce on the tree (smallest derivation of the same type), localize to a plan setting, rewrite or version.
- **LLM** reads recent commits and proposes grammar; a **gate** on the server accepts it only if it runs and executes the commit's changed lines.

Arbor: all eleven bugs and how they were found

Bug type	Found by	Signal	Upstream
assertion (crash)	LLM fragment + TLP check	assertion during a metamorphic check	**fixed** #19803
wrong output	cross-version oracle	master and 18.6 disagreed	**fixed, our patch** #19771
assertion (crash)	extended-statistics generator	assertion in the estimator	**fixed, our patch**
wrong result + assertion	grammar fuzzer + cross-version	assertion on master, wrong answer on 18.6	**fixed, our patch** #19768
wrong result	cross-version + TLP	extra row on master	**fixed, our patch**
hang	hang oracle	timeout and cancel ignored	**fixed, our patch**
internal error	internal-error oracle (`XX000`)	“should not happen” error	**fixed** #19802
wrong result	$e \rightarrow (\mathrm{SELECT}\ e)$ rewrite	sub-select blocked a simplification	**open** #19793
bad estimate	CERT oracle	estimate rose under a restriction	**open** (patch)
wrong result	plan-variant oracle	NULL serially, 1 in parallel (`corr()`)	**fixed, our patch** #19826
internal error	internal-error oracle on newer master	Memoize “subplan’s variable” error	**fixed** #19828

11 reported, 9 fixed upstream (6 with our patches), 2 under review. The tool rediscovers the first 9 in one 150-second run.

Arbor: three bugs in detail

#19768 `JSON_TABLE`

State: fixed

Type: wrong result in released 17/18, assertion (crash) on master.

What happened: `JSON_TABLE` never assigned collations to its input, so a row comparison inside it was compiled with *zero* column comparisons.

How found: the grammar fuzzer hit the assertion on master; the cross-version oracle showed 18.6 silently answering wrong.

Fix: our patch, committed by Amit Langote, back-patched to 17.

`NOT IN` anti-join

State: fixed

Type: silent wrong result (extra row) on master.

What happened: a new optimization turns `NOT IN` into an anti-join, valid only if no value can be NULL; an implicit cast *can* produce NULL, so a row that must be filtered out was returned.

How found: cross-version (18.6 has no such optimization) and TLP (the partitions did not add up).

Fix: our patch, committed by Richard Guo.

#19803 parallel `Append`

State: fixed

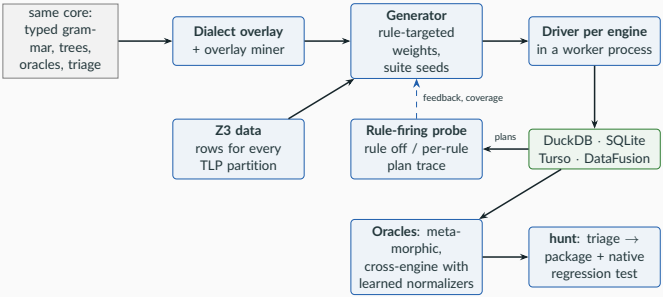
Type: assertion failure (server crash).

What happened: the planner built a parallel plan, later found a child not safe for parallel execution, but kept the parallel plan; an internal assertion stopped the server.

How found: the LLM read a recent planner commit and added grammar for its feature; a derived query crashed the server during a TLP check; the assertion came from the server log; triage reduced it.

Fix: David Rowley, back-patched to 14.

Generalized fuzzer: architecture and techniques



- **Driver per engine** in a worker process (DuckDB, SQLite, Turso, DataFusion); same trees, oracles, triage.
- **Dialect overlays** on the same grammar + **overlay miner** (79–98% valid).
- **Optimizer-rule firing** instead of code coverage: feedback, coverage, rule-targeted generation.
- **Z3 data synthesis**: rows for every TLP partition.
- **Learned normalizers**: cross-engine comparison (disagreements 28.7% $\rightarrow$ 3.6%).
- **Autonomous loop**: campaign $\rightarrow$ triage $\rightarrow$ package with a regression test in the engine's own format.

Generalized fuzzer: three bugs in detail

Turso

State: filed with fix

Type: wrong result / spurious error.

What happened: `percentile_disc/percentile_cont` read their arguments from consecutive registers, but an expression argument was compiled into a temporary register, so the aggregate read the wrong value.

How found: Turso-vs-SQLite differential oracle; reduced to one aggregate call.

Fix: arguments compiled into one register block. Issue #11594, PR #11595 (12,700 tests pass).

DuckDB

State: approved by a maintainer

Type: internal error, regression in 2.0.0.

What happened: for `min/max` over a nested value the binder copied the argument before checking whether a collation applies; an argument containing a subquery cannot be copied.

How found: internal-error oracle on the nightly build.

Fix: copy only when a collation applies. Issue #28551, PR #28552 (collation and subquery suites pass).

DataFusion

State: filed with fix

Type: internal error in `UPDATE/DELETE`.

What happened: common-subexpression elimination replaced a repeated expression by a column that exists only in the plan; DML then asked the table for that column.

How found: internal-error oracle; re-check on `main` found the `WHERE` clause fails too.

Fix: inline such columns before DML. Issue #27235, PR #27236 (all 546 test files pass).

Other bugs found: a DataFusion internal error with subqueries in `VALUES` and a Turso panic with an aggregate in `IN (SELECT ...)` next to a window function (both with a fix ready); a Turso panic, a rejected Turso join and a DuckDB bind error, all three already fixed upstream when we re-checked on the newest code (`main`).

Results and conclusion

11
PostgreSQL bugs reported

9
fixed upstream (6 our patches)

3
fixes filed to DuckDB, Turso, DataFusion

36.7%
backend lines (random 22.2%)

- Evaluation: 5 trials $\times$ 1 h per arm; 93 candidates triaged; 23 re-injected real bugs as ground truth.
- Lesson: valid SQL is the easy part; oracle precision decides how useful a fuzzer is.
- Team: Venu (tool, paper, generalized fuzzer) · Bhuvan (benchmark, bug hunts, four patches) · Gowtham (campaigns, LLM arm, two reports, one fix).

Thank you. Questions?

github.com/PASS-IITG/cs5187-term-project-blazing-bhuvi